

Lecture 15

Tuesday March 31 2026

Review

- Project 1 is due Thursday one student submits with screen shot of the program working
- Project 2 will be released
- Semaphores have two functions
 - `wait()` locks the resource
 - `signal()` releases the resource
- Look over parking lot problem example

Bounded-Buffer Problem

- Bounded buffer uses a producer/consumer
- if the consumer accesses the queue before the producer the producers will not be able to access it.
- Solution is to lock the queue with the full/empty variables outside to provide better control over who has access

Readers and Writers Problem

- Need to have synchronization between two people who are trying to read and write to a file at the same time
- Two readers are not an issue
- If a writer is working on the file the reader should not be able to access it.
- Two writers are also an issue because they are changing the same thing
- The lock needs to focus on the writer to stop them from accessing the file when readers have access to it but all readers can still access it
- need to create a variable for reader count
- The first reader locks the write lock and the last reader opens it for the writers
- `reader_count == 1` means first reader -> locks writers lock
- `reader_count == 0` means last reader -> opens readers lock
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